

Dr Milan Župunski

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RESEARCH PROFILE

Plant cell biology and live-cell imaging of signal perception in roots. Quantitative spatiotemporal analysis of calcium and small-molecule biosensor responses, and a reference atlas of elicitor signatures in *Arabidopsis* roots. Current focus: how the perception of biotic and abiotic cues changes through time.

EDUCATION

- 2017 Ph.D., Ecology, Faculty of Sciences, University of Novi Sad, Novi Sad, Serbia
- 2010 M.Sc., Biology, Faculty of Sciences, University of Novi Sad, Novi Sad, Serbia
- 2009 B.Sc. (Diploma), Biology, Faculty of Sciences, University of Novi Sad, Novi Sad, Serbia

ACADEMIC TITLE

- 2026 Senior Research Associate (Viši naučni saradnik), Ministry of Science, Technological Development and Innovation, Republic of Serbia; research-track equivalent of associate professor (decision no. 660-01-00001/2026-03/215, 13.07.2026)

PROFESSIONAL APPOINTMENTS

- 01.03.2026– present Independent research and manuscript preparation, Novi Sad, Serbia (research from HHU Düsseldorf and ISTA)
- 01.10.2025– 28.02.2026 Researcher, Group of Prof. Dr. Jiří Friml, Institute of Science and Technology Austria (ISTA), Klosterneuburg, Austria
- 01.11.2021– 30.09.2025 Researcher, Institute of Cell and Interaction Biology, Heinrich Heine University Düsseldorf (HHU), Germany
- 01.01.2020– 31.10.2024 Docent, Faculty of Sciences, University of Novi Sad, Novi Sad, Serbia (on leave of absence 2021–2024 for research at HHU)
- 01.01.2014– 31.12.2019 Teaching Assistant, Faculty of Sciences, University of Novi Sad, Novi Sad, Serbia
- 01.07.2012– 31.12.2013 Research Trainee, Faculty of Sciences, University of Novi Sad, Novi Sad, Serbia

PUBLICATIONS

h-index 18, 1275 citations (Google Scholar, October 2026)

Plant Cell Biology and Signaling

Župunski M, Brugman R, Zauser M, Lampou K, Waadt R, Guichard M, Krebs M, Schumacher K, Lamar R, Pahle J, Grossmann G. (2026). A calcium-fingerprint atlas infers the signaling modes of complex elicitors and reveals a salt-like response to a humic biostimulant. *bioRxiv*, 756665.

- Župunski M**, Grossmann G, and Friml J. (2026). Decoding dialogue: The auxin–calcium signaling crosstalk in plants. *Trends in Plant Science*, <https://doi.org/10.1016/j.tplants.2026.08.002>.
- Erdrich S, Keilhammer M, Sharma S, **Župunski M**, Schurr U, Grossmann G, Frunzke J, Arsova B. (2026). Phage biocontrol reduces the disease burden and modulates plant immunity through suppression of bacterial virulence. *Cell Reports*, 45(7): 117622. <https://doi.org/10.1016/j.celrep.2026.117622>
- Gilliard G, Pršić J, Crowet J-M, Chemotti C, Ahmed J, Lorent J, Jolivet M-D, Egli S, Egée S, Bouyer G, Race G, van Buren L, Van Den Berghe A, Argüelles-Arias A, Mathelié-Guinlet M, Ibrahim H, Genva M, Fouillen L, Mirande-Bret C, Razin O, Vue B, **Župunski M**, Luzuriaga-Loaiza WP, Deboever E, Nasir MN, Lins L, Van Der Smissen P, Boubsi F, Eschrig S, Germain V, Höfte M, Zipfel C, Dufrêne YF, Tyteca D, Koutsoubas A, Brocca P, Grossmann G, Ranf S, Dorey S, De Coninck B, Nürnberger T, Mongrand S, Gronnier J, Rondelli V, Deleu M, Ongena M. (2026). Membrane remodelling mediates lipopeptide-induced immunity in Arabidopsis. *Nature Plants*, 12: 1034–1050. <https://doi.org/10.1038/s41477-026-02270-3>
- Babić D, **Župunski M**, Friml J. (2026). Imaging and genetic toolbox to study Arabidopsis embryogenesis. *New Phytologist*, 250: 1483–1491. <https://doi.org/10.1111/nph.71072>
- Santos MP, **Župunski M**, Monda H, et al. (2025). Humic acids modify root architecture in Arabidopsis through H⁺-ATPase-dependent target of rapamycin activation in concert with Ca²⁺ and ROS signaling. *Chemical and Biological Technologies in Agriculture*, 12: 47. <https://doi.org/10.1186/s40538-025-00764-4>
- Chaudhary A, Hsiao Y-C, Yeh F-L J, **Župunski M**, Zhang H, Aizezi Y, Malkovskiy A, Grossmann G, Wu H-M, Cheung AY, Xu S-L, Wang Z-Y. (2025). FERONIA signaling maintains cell wall integrity during brassinosteroid-induced cell expansion in Arabidopsis. *Molecular Plant*, 18(4): 603–618. <https://doi.org/10.1016/j.molp.2025.02.001>
- VanBuren R, Wai CM, Giarola V, **Župunski M**, Pardo J, Kalinowski M, Grossmann G, Bartels D. (2023). Core cellular and tissue-specific mechanisms enable desiccation tolerance in Craterostigma. *Plant Journal*, 114: 231–245. <https://doi.org/10.1111/tpj.16165>
- Sadoine M, De Michele R, **Župunski M**, Grossmann G, Castro-Rodriguez V. (2023). Monitoring nutrients in plants with genetically encoded sensors: achievements and perspectives. *Plant Physiology*, kiad337. <https://doi.org/10.1093/plphys/kiad337>

Preprints

- Fuchs VAF, Denninger P, **Župunski M**, Jaillais Y, Engel U, Grossmann G. (2021). Nanodomain-mediated lateral sorting drives polarization of the small GTPase ROP2 in the plasma membrane of root hair cells. *bioRxiv*, 459822. <https://doi.org/10.1101/2021.09.10.459822>

Plant Ecophysiology

- Matijević I, Arsenov D, Popov N, Živkov-Baloš M, Nikolić N, Pajević S, Borišev M, **Župunski M**. (2025). Acclimatization mechanisms in *Populus deltoides* roots to zinc excess. *Acta Physiologiae Plantarum*, 47: 103. <https://doi.org/10.1007/s11738-025-03850-0>
- Joksimović A, Arsenov D, Borišev M, Djordjević A, **Župunski M**, Borišev I. (2025). Foliar application of fullereneol and zinc oxide nanoparticles improves stress resilience in drought-sensitive *Arabidopsis thaliana*. *PLoS ONE*, 20: e0330022. <https://doi.org/10.1371/journal.pone.0330022>
- Borišev M, **Župunski M***, Arsenov D, Nikolić N, Tarčak S, Pajević S. (2024). Understanding beech (*Fagus sylvatica* L.) photosynthetic responses to microhabitat water deficit: a site-specific investigation. *European Journal of Forest Research*. <https://doi.org/10.1007/s10342-024-01702-z>
- Beljin J, Arsenov D, Slijepčević N, Maletić S, Đukanović N, Chalot M, **Župunski M**, Tomašević Pilipović D. (2023). Recycling of polluted dredged sediment – building new materials for plant growing. *Waste Management*, 166: 13–24.
- Nikolić N, Pajević S, Arsenov D, Borišev M, **Župunski M**. (2023). Breaking the myth of healthy food production in rural areas: cases studied in Vojvodina Province (Serbia). *Environmental Science and Pollution Research*, 30: 4778–4791. <https://doi.org/10.1007/s11356-022-22466-2>
- Nedić M, **Župunski M**, Orlović S, Kovačević B, Kebert M, Vaštag E, Miljković D, Gutalj M, Markić A, Stojnić S. (2023). Assessment of the phenotypic diversity of wild cherry (*Prunus avium* L.) populations and halfsib lines by multivariate statistical analyses. *Silvae Genetica*, 71(1): 116–127. <https://doi.org/10.2478/sg-2022-0014>

- Župunski MŠ**, Arsenov D, Borišev M, Nikolić N, Pajević S. (2021). Should I GROW or should I SLOW: a meta-analysis of fast-growing tree species grown in cadmium-perturbed environments. *Physiologia Plantarum*, 174: e13594. <https://doi.org/10.1111/ppl.13594>
- Arsenov D, **Župunski M**, Pajević S, Nemeš I, Simin N, Alnuqaydan AM, Watson M, Aloliqi AA, Mimica-Dukić N. (2021). Roots of *Apium graveolens* and *Petroselinum crispum* – insight into phenolic status against toxicity level of trace elements. *Plants*, 10(9): 1785. <https://doi.org/10.3390/plants10091785>
- Arsenov D, **Župunski M**, Pajević S, Borišev M, Nikolić N, Mimica-Dukić N. (2021). Health assessment of medicinal herbs, celery and parsley related to cadmium soil pollution – potentially toxic elements (PTEs) accumulation, tolerance capacity and antioxidative response. *Environmental Geochemistry and Health*, 43: 2927–2943. <https://doi.org/10.1007/s10653-020-00805-x>
- Lazić D, Putnik-Delić M, Daničić M, **Župunski M**, Arsenov D, Vuković S, Maksimović I. (2020). Efficiency of Si in alleviating NaCl-induced stress in oilseed rape. *Pakistan Journal of Agricultural Sciences*, 57(4): 901–907.
- Arsenov D, **Župunski M**, Borišev M, Nikolić N, Pilipović A, Orlović S, Kebert M, Pajević S. (2020). Citric acid as soil amendment in cadmium removal by *Salix viminalis* L., alterations on biometric attributes and photosynthesis. *International Journal of Phytoremediation*, 22(1): 29–39. <https://doi.org/10.1080/15226514.2019.1633999>
- Lalić D, Meriluoto J, Zorić M, Dulić T, Miroslavljević M, **Župunski M**, Svirčev Z. (2020). Potential of cyanobacterial secondary metabolites as biomarkers for paleoclimate reconstruction. *CATENA*, 185: 104283. <https://doi.org/10.1016/j.catena.2019.104283>
- Arsenov D, Nikolić N, Borišev D, **Župunski M**, Orlović S, Pilipović A, Pajević S. (2019). Greenhouse assessment of citric acid-assisted phytoremediation of cadmium by willows (*Salix* spp.) – effect on photosynthetic performances and metal tolerance. *Baltic Forestry*, 25(2): 203–212.
- Horak R, **Župunski MŠ***, Pajević S, Borišev M, Arsenov D, Nikolić N, Orlović S. (2019). Carbon assimilation in oak (*Quercus* spp.) populations under acute and chronic high-temperature stress. *Photosynthetica*, 57(3): 875–889.
- Pajević S, Arsenov D, Nikolić N, Borišev M, Orčić D, **Župunski M**, Mimica-Dukić N. (2018). Heavy metal accumulation in vegetable species and health risk assessment in Serbia. *Environmental Monitoring and Assessment*, 190(8): 459.
- Prakash Mishra A, Sharifi-Rad M, Ali Shariati M, Mabkhot YN, Al-Showiman SS, Rauf A, Salehi B, **Župunski M**, Sharifi-Rad M, Gusain P, Sharifi-Rad J, Suleria HAR, Iriti M. (2018). Bioactive compounds and health benefits of edible *Rumex* species – a review. *Cellular and Molecular Biology*, 64(8): 27–34.
- Arsenov D, **Župunski M**, Borišev M, Nikolić N, Orlović S, Pilipović A, Pajević S. (2017). Exogenously applied citric acid enhances antioxidant defense and phytoextraction of cadmium by willows (*Salix* spp.). *Water, Air, & Soil Pollution*, 228: 221. <https://doi.org/10.1007/s11270-017-3405-6>
- Bojović M, Nikolić N, Borišev M, Pajević S, **Župunski M**, Horak R, Pilipović A, Stojnić S. (2017). The diurnal time course of leaf gas exchange parameters of pedunculate oak seedlings subjected to experimental drought conditions. *Baltic Forestry*, 23(3): 584–594.
- Borišev M, Borišev I, **Župunski M**, Arsenov D, Pajević S, Ćurčić Ž, Vasin J, Djordjević A. (2016). Drought impact is alleviated in sugar beets (*Beta vulgaris* L.) by foliar application of fullerene nanoparticles. *PLOS ONE*, 11(11): e0166248.
- Borišev M, Pajević S, Nikolić N, Orlović S, **Župunski M**, Pilipović A, Kebert M. (2016). Magnesium and iron deficiencies alter Cd accumulation in *Salix viminalis* L. *International Journal of Phytoremediation*, 18(2): 164–170.
- Župunski MŠ**, Borišev M, Orlović S, Arsenov D, Nikolić N, Pilipović A, Pajević S. (2016). Hydroponic screening of black locust families for heavy metal tolerance and accumulation. *International Journal of Phytoremediation*, 18(6): 583–591.
- Nikolić N, Borišev M, Pajević S, Arsenov D, **Župunski M**, Orlović S, Pilipović A. (2015). Photosynthetic response and tolerance of three willow species to cadmium exposure in hydroponic culture. *Archive of Biological Sciences*, 67(4): 1411–1420.

- Borišev M, Pajević S, Orlović S, Nikolić N, **Župunski M**, Pilipović A. (2015). Daily dynamics of photosynthetic parameters in beech population under periodical drought conditions. *Open Life Sciences*, 10: 165–174.
- Nikolić N, Borišev M, Pajević S, **Župunski M**, Topić M, Arsenov D. (2014). Responses of wheat (*Triticum aestivum* L.) and maize (*Zea mays* L.) plants to cadmium toxicity in relation to magnesium nutrition. *Acta Botanica Croatica*, 73(2): 359–373.
- Topić M, Borišev M, Orlović S, Tomičić M, **Župunski M**, Nikolić N, Pajević S, Krstić B, Pilipović A. (2013). Clonal differences of black poplar cuttings for morpho-physiological and biochemical responses to soil water deficits. *Journal of Animal and Plant Sciences*, 23(6): 1725–1732.
- * / § Corresponding author (as designated in the original publication).

Book Chapters

- Borišev I, Borišev M, Jović D, **Župunski M**, Arsenov D, Pajević S, Djordjević A. (2020). Nanotechnology and remediation of agrochemicals. In: *Agrochemicals Detection, Treatment and Remediation – Pesticides and Chemical Fertilizers* (Prasad MNV, ed.). Butterworth-Heinemann, Elsevier, pp. 487–533.
- Župunski M**§, Pajević S, Arsenov D, Nikolić N, Pilipović A, Borišev M. (2018). Insights and lessons learned from the long-term rehabilitation of abandoned mine lands – a plant-based approach. In: *Bio-Geotechnologies for Mine Site Rehabilitation* (Prasad MNV, Favas PJC, Maiti SK, eds.). Elsevier, pp. 215–232.
- Borišev M, Pajević S, Nikolić N, Pilipović A, Arsenov D, **Župunski M**. (2018). Mine site restoration using silvicultural approach. In: *Bio-Geotechnologies for Mine Site Rehabilitation* (Prasad MNV, Favas PJC, Maiti SK, eds.). Elsevier, pp. 115–130.
- Pajević S, Borišev M, Nikolić N, Arsenov D, Orlović S, **Župunski M**§. (2016). Phytoextraction of heavy metals by fast-growing trees: a review. In: *Phytoremediation: Management of Environmental Contaminants*, vol. 3 (Ansari AA, Gill SS, Gill R, Lanza GR, Newman L, eds.). Springer, pp. 29–64.

* / § Corresponding author (as designated in the original publication).

Textbooks and Practicums

- "Instrumental Methods in Biology" (Инструменталне методе у биологији), co-author, University of Novi Sad, Serbia. 2021. ISBN 978-86-7031-539-6
- "Fundamentals of Plant Physiology – Practicum for Ecology Students" (Основи Физиологије Биљака), co-author, University of Novi Sad, Serbia. 2014. ISBN 978-86-7031-328-6

RESEARCH INFRASTRUCTURE AND CORE FACILITY EXPERIENCE

2021–2025 Instrument responsibility – Institute of Cell and Interaction Biology, HHU Düsseldorf

Day-to-day responsibility for the institute's three light microscopy systems (spinning-disk confocal, high-sensitivity epifluorescence, fluorescence stereomicroscope), serving approximately 15 registered users at 6–8 hours of daily use per system. User management, training, access and scheduling; procurement of components; support during installation; calibration and light-path alignment; regular performance checks; preparation of written standard operating procedures; troubleshooting; direct contact with manufacturers on service and repair.

2021–2025 DFG Major Research Instrumentation Program (Großgeräteantrag) – robotic kinetic-phenotyping microscopy platform, €1.5 million (funded; PI: Prof. Dr. Guido Grossmann)

Contributed to the scientific concept and imaging workflow; drafting of proposal sections; collection and comparison of equipment quotations for the costing; contribution to installation planning. Platform built and entering routine operation.

Facility training

2019 "Imaging Cells, Organoids and Organs," Course of the Nikon Imaging Center at the University of Heidelberg (25–28 March 2019)

Image data management and analysis

OMERO for image data management and metadata handling in line with FAIR principles. Quantitative image analysis in FIJI with custom macro scripting, and in Python. Co-developer of *Crestline*, with Prof. Dr. Guido Grossmann, a pipeline for quantitative spatiotemporal analysis of biosensor time-series data (manuscript in preparation).

Sample preparation development: designed RootComb, a 3D-printed imaging chamber insert for applying treatments to growing roots during live imaging and recovering imaged tissue for fixation within seconds (manuscript in preparation).

Microscopy techniques

Epifluorescence, including open-frame modular systems (Cairn Research); spinning-disk confocal (Nikon Eclipse Ti / CrestOptics); laser-scanning confocal (Zeiss LSM 800/900); vertical-stage confocal imaging of roots (Zeiss LSM 800); fluorescence stereomicroscopy (Olympus MVX10); ratio and biosensor imaging with genetically encoded sensors and reporter lines; brightfield microscopy; scanning electron microscopy (JEOL JSM-6460LV, teaching).

RESEARCH VISITS AND FELLOWSHIPS

2019–2020 Guest Scientist, Group of Dr. Guido Grossmann, Centre for Organismal Studies (COS), University of Heidelberg, Germany – calcium signaling in plant–environment interactions (Jan–Apr 2019; Nov 2019–Mar 2020; 8 months total)

INVITED TALKS AND TEACHING VISITS

- 2024 "Decoding Environmental Heterogeneity Through Orchestrated Cell–Cell Communication." Invited seminar talk, Institute of Science and Technology Austria (ISTA), Klosterneuburg, Austria (host: Prof. Dr. Jiří Friml, December 2024)
- 2019 Invited teaching and research visit, University of Heidelberg, Germany – lecture and workshop series for PhD students and early-career researchers (HBIGS graduate program); Erasmus+ Staff Mobility (KA107), 3 weeks (April 2019)
- 2018 Invited teaching visit, University of Eastern Finland – lecture and workshop series on "Phytoremediation" for PhD students and early-career researchers (8 contact hours); Erasmus+ Staff Mobility (KA1), 1 week (March 2018)

CONFERENCE PRESENTATIONS

- 2024 "Decoding Environmental Signals: Calcium Responses to Abiotic and Biotic Stimuli in Arabidopsis Roots." The 5th International Conference on Plant Biology (24th SPSS Meeting), Srebrno Jezero, Serbia. (Selected talk)
- 2022 Župunski M, Brugman R, Zauser M, Löffler J, Pahle J, Grossmann G. "Semi-automated spatiotemporal quantification of calcium responses in *A. thaliana* roots." Plant Calcium Signalling, Università degli Studi di Milano, Milan, Italy, p. 45. (Poster)
- 2018 "Optimization of Cell Redox Status by Fullerenol Nanoparticles Seed Priming in *Brassica napus* Plants." The International Bioscience Conference and the 7th Joint International PSU-UNS Bioscience Conference, Krabi, Thailand. (Invited talk)

AWARDS AND GRANTS

- 2022 Postdoctoral Research Fellowship, Ministry of Education, Science and Technological Development, Republic of Serbia (XVIII National Call for Postdoctoral Training Abroad) – 6-month fellowship (€4,200) for research training at the Institute of Cell and Interaction Biology, Heinrich Heine University Düsseldorf, Germany (Apr–Oct 2022)

Curriculum Vitae

2019 Travelling Fellowship, The Company of Biologists / Journal of Cell Science (€1,901) – research visit to the Grossmann Lab, University of Heidelberg, Germany (Oct–Dec 2019)

PATENTS AND TECHNOLOGY TRANSFER

2024 National Patent (RS 66188 B1, Republic of Serbia) and 4 registered technical solutions, developed from collaborative research on carbon-nanomaterial (fullerenol) biostimulant; co-founder, AquaBoostNano, Novi Sad, Serbia (until 2026), startup translating carbon-nanomaterial research into applied biostimulant products

TEACHING EXPERIENCE

University of Novi Sad, Faculty of Science, Department of Biology and Ecology

Plant Physiology (undergraduate) (2012–2021)

Instrumental Analysis in Biology (undergraduate) (2012–2021); light and scanning electron microscopy; sample preparation and imaging on a JEOL JSM-6460LV

University of Heidelberg, HBIGS

Principal Component Analysis in Biosciences, Core Course (2019)

Heinrich Heine University Düsseldorf (HHU)

"Challenges and Opportunities of Translational Developmental Plant Biology: From Model to Crop," B.Sc./M.Sc. module (2023–2025)

STUDENT SUPERVISION AND MENTORING

University of Novi Sad, Faculty of Science

Master's Thesis – Nikola Radovanović, "Photosynthetic Efficiency of Black Locust (*Robinia pseudoacacia* L.) Grown on Soil Contaminated with Heavy Metals" (2020)

Bachelor's (Diploma) Thesis – Adam Miličić, "The Importance of Silviculture in the Restoration and Rehabilitation of Land Polluted by Heavy Metals" (2021)

PhD Dissertation Committee Member – Milena Stanković-Nedić, "Variability of Phenotypic Traits of Wild Cherry (*Prunus avium* L.) as a Basis for Genetic Resource Conservation," Faculty of Agriculture, University of Novi Sad (defended September 2024)

Heinrich Heine University Düsseldorf (HHU)

M.Sc. Student Co-Mentoring – Anna Thimm (2024) (daily supervision; official advisor: Prof. Dr. Guido Grossmann)

Undergraduate Research Rotation Mentoring – Maximilian Opitz (2023); Digila Elizabeth Francis (2025)

Thesis Advisory Committee (TAC) External Supervisor – Christian Frederic Kaiser, PhD candidate, HHU

SERVICE TO THE PROFESSION

Journal Peer Review – BMC Plant Biology, Plant Cell Reports, Frontiers in Plant Science, Photosynthetica, Scientific Reports, BioMetals, Chemosphere, Ecotoxicology and Environmental Safety, European Journal of Forest Research, Food Chemistry, Journal of Hazardous Materials, International Journal of Phytoremediation, Journal of Soils and Sediments, Journal of Soil Science and Plant Nutrition (24 verified reviews; Web of Science Researcher ID: U-1696-2017)

Grant Review – National Science Centre (NCN), Poland (full-length proposal)

COMMUNITY INVOLVEMENT AND OUTREACH

2011-2019 Workshops for public science engagement, "Night of Biology" (annual)

RELATED PROFESSIONAL SKILLS

Wet lab: molecular biology and biochemistry assays; CRISPR-Cas9 construct design and generation of edited Arabidopsis lines; microfluidics fabrication and experimentation

Curriculum Vitae

Computational: R (tidyverse, ggplot2, RMarkdown); Python (pandas, matplotlib); RNA-seq analysis (DESeq2, edgeR, limma); statistics (exploratory and inferential methods, multivariate analysis, random forest machine learning)

LANGUAGES

Serbian: native

English: fluent; working language for research and teaching since 2019

German: basic (A2)

PROFESSIONAL MEMBERSHIPS

Serbian Society for Plant Physiology (SPPS), Federation of European Societies of Plant Biology (FESPB), Society for Experimental Biology (SEB)

REFERENCES

Available on request.